

Practice Problems and Assignment

Lecture 1 and 2:

1. **(Mid)** Calculate the degree sum in a complete graph of 5 vertices. Derive a general formula for calculating the degree sum of a complete graph with “n” vertices.
2. **(Mid)** What is the highest size of clique you can get in W_8 ?
3. **(Mid)** Draw P_4 , C_5 and W_7
4. Prove that W_n doesn't have an induced K_4 for $n > 4$.
5. **(Mid)** Prove that number of odd degree vertices is even.
6. **(Mid)** Derive the formula for the number of edges for W_n , C_n , P_n .

Lecture 3:

1. **(Mid)** Prove that, “For any graph of seven vertices, G or G' contains an induced K_3 ”
2. **(Mid)** Prove that, “If a graph G is a complete graph, then it can not contain an induced P_3 ”

Lecture 4:

1. **(Mid)** 7,7,6,6,4,1 is this a valid degree sequence? Give proper reasoning.
2. **(Mid)** 5,5,3,1 Is this a valid degree sequence? If yes, then draw a graph (possibly a multigraph) with this degree sequence.
3. **(Mid)** Is 7,4,3,3,2,1,1,1 a graphic sequence? If yes, then draw a simple graph with this sequence.
4. **(Mid)** Show the adjacency matrix and adjacency list representation of the above graph.
5. **(Mid)** Suppose, you have a cycle graph of 6 vertices. Show the adjacency list representation of that graph.
6. **(Mid)** Let's say you have an adjacency list representation of a cycle graph C_n . What is the space complexity? What is the time complexity of checking an edge between two vertices u and v . Write the best possible answer.

Lecture 5:

1. **(Mid)** You have 6 vertices. What is the possible number of simple undirected graphs with those vertices? What is the possible number of simple directed graphs with those vertices?
2. **(Mid)** Let's say you want to add an additional constraint on problem 1. Among the vertices, the sixth vertex can only have outgoing edges. Then how many simple directed graphs are possible?
3. **(Mid)** Draw the complement graph of $K_{3 \times 4}$. What is the clique of the maximum size of that graph?

Lecture 6:

1. **(Mid)** Is it possible to have a path that is not a trail? Give proper reasoning.
2. **(Mid)** Let a graph G be a disconnected graph. Show that the complement of G is a connected graph.
3. **(Mid)** You have taught your friend about the lemma "*Every graph with n vertices and m edges has at least $n - m$ connected components*". Then your friend says the following, "*If I have a complete graph with 4 vertices then that graph has 6 edges. According to the lemma it means I have $4 - 6 = -2$ connected components which is clearly false, Thus the lemma is false*". Explain what is wrong with the above argument.
4. **(Mid)** What is the necessary and sufficient condition of the following statement "An edge is a cut-edge if and only if it belongs to no cycle". ?
5. **(Mid)** How many cut edges does P_n have?
6. **(Mid)** Prove that, "*If there is a circuit starting and ending with vertex "x" then there is a cycle starting and ending with vertex "x".*"
7. **(Mid)** Prove the necessary condition of, "*A graph G is bipartite if and only if G doesn't contain any odd cycle*".

Lecture 7, 8, 9:

1. **(Mid)** Write the necessary and sufficient statement of "*A connected graph G is Eulerian if and only if every vertex of G has even degree*".
2. **(Mid)** Prove that, "*A connected graph G is Eulerian if and only if every vertex of G has even degree*".
3. To practice the cycle decomposition of Eulerian graphs and finding Eulerian circuits in a Eulerian graph, just draw a graph with all even degree vertices. Then try to solve for that graph.

4. **(Mid)** Prove the sufficient condition of, "*A graph G is bipartite if and only if G doesn't contain any odd cycle*".

Lecture 10:

1. **(Mid)** For a complete graph K_n to be Hamiltonian and Eulerian, what is the condition that needs to be imposed on " n "? (A graph is hamiltonian if the graph has a hamiltonian cycle.)
2. **(Mid)** Find two edge disjoint hamiltonian cycles in K_5
3. **(Mid)** Prove that, "Let G be a connected simple graph. Then $\kappa'(G) \leq \delta(G)$ ".
4. **(Mid)** Find $\kappa(G)$, $\kappa'(G)$ and $\delta(G)$ of a wheel graph W_n .

Lecture 11:

1. Prove that, "Let G be a connected simple graph. Then $\kappa(G) \leq \delta(G)$ ".

Lecture 12:

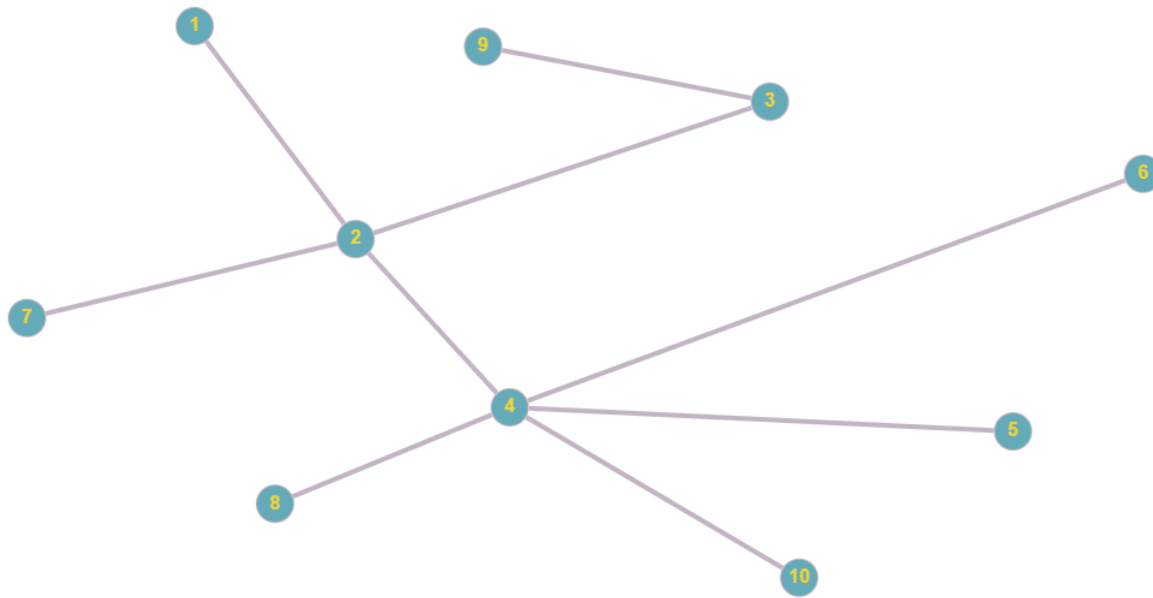
1. **(Final)** Draw a complete binary tree that is also a full binary tree.
2. **(Final)** Prove that, "The number of leaves in a full binary tree is one more than the number of internal vertices."
3. **(Final)** Is it possible to draw a full binary tree with 100 vertices? Give proper justification.
4. **(Final)** Let's take a full trinary tree where each vertex is either a leaf or has 3 children. Can you come up with a relation between internal vertices and leaves in this case?

Lecture 13:

1. **(Final)** Prove that, If G be a graph with n vertices, then, any two of the following three statements imply the third (and characterize a tree of n vertices).

(a) G is connected. (b) G contains no cycle. (c) G has $n - 1$ edges.

Lecture 14:



1. **(Final)** Construct a unique sequence from the above tree.
2. **(Final)** Reverse the sequence that you got in question 1. Then construct a unique tree from that sequence. Is it the same tree as above or a different one?
3. **(Final)** While constructing a tree from a sequence, how are we ensuring that the resulting graph contains no cycle?

Lecture 15:

1. **(Final)** For the above graph, make a rooted tree using 3 as a root. Then, find the maximum independent set in that tree.

Lecture 16 and 17:

1. **(Final)** Prove the following statement by induction, "Let G be a connected plane graph, and let n , m , and f denote, respectively, the numbers of vertices, edges, and faces of G . Then $n - m + f = 2$."
2. **(Final)** Prove that, "If G is a planar graph with $n(\geq 3)$ vertices and m edges, then, $m \leq 3n - 6$ "
3. **(Final)** Prove that, "Let G be a planar graph. Then G has at least one vertex of degree at most 5"
4. **(Final)** A planar connected graph has 15 vertices. What is the range for the number of edges?
5. Show that, $m \leq 2n - 4$ for a planar bipartite graph of n vertices and m edges.

